CS 460
Computer Vision
Lecture 6

Spring, 2026

February

CALENDAR

February 5th - Committee Showcase

February 12th - Molly Lodge - US Coast Guard

February 19th - Kelly Tallis & Johanna Urena - Spencer's Gifts

February 26th - Bruce Katz- RunSignup

Weekly Meetings Thursday
6:30PM in Robinson 311

WEBSITE

INSTAGRAM

DISCORD

Lockheed Martin x Rowan Information Session

Join us for an on-campus event giving students a deeper look inside Lockheed Martin. Grab a bite to eat and gain insight into who we are and what we do!

Topics: Career Development, Company Culture / Work-life Balance and more!

Date: March 5th 2026

Time: 5:00 – 6:30 p.m

Location: Engineering Hall 140-141

Targeted Majors include:
Electrical and Computer Engineering
Electrical Engineering Technology
Mechanical Engineering
Mechanical Engineering Technology
Computer Science
& related

Announcements

Rowan Career Fairs

https://rowan-csm.symplicity.com/events/19de426e7760a9f73cb50047a00da60b/overview

Announcement

Undergrad Midterm exam presentations

3/10/26

Grad Midterm exam presentations

3/13/26

Make sure you are signed up on the class sign-in sheet for a date and topic

Topic list to go up on canvas soon!

Or you can suggest one!

Agenda

Pre-trained models

Transfer Learning

tfjs pre-trained model

COCO_SSD

Agenda

tfjs pre-trained model

tfjs Pre-Trained models

- Before we build a model from scratch is important to be aware that there may exist technologies that can be leveraged
- Pre-Trained Models
  - Models that have already been trained
- Eliminates data gathering burden
- Eliminates training step
- You can load the pre-trained model and use it for the problem domain it was built for

tfjs Pre-Trained models

- Two forms
  - Type1 are wrapped into easy to JS classes
    - Can be used by creating a few lines of JS code
    - Can help solve many typical CV problems
    - Can be useful for Machine Learning novices
  - Type2
    - Require more knowledge of Machine Learning
    - Available in raw form
    - No helper functions

tfjs Pre-Trained models

- Object Detection (coco-ssd)
  - Detecting objects in image and specifically locating each object in the image
  - Microsoft's Common Objects in Context (COCO) dataset
  - Uses the Single Shot MultiBox Detection (SSD) architecture
  - Model is trained to recognize up to 80 everyday common objects
  - Draw bounding box with labels
- https://github.com/tensorflow/tfjs-models/tree/master/coco-ssd#readme
- Multiple Object Detection

tfjs Pre-Trained models

- Object Detection (coco-ssd)

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>COCO-SSD Demo</title>
  <!-- Load TensorFlow.js -->
  <script src="https://cdn.jsdelivr.net/npm/@tensorflow/tfjs"></script>
  <!-- Load the coco-ssd model -->
  <script src="https://cdn.jsdelivr.net/npm/@tensorflow-models/coco-
   ssd"></script>
</head>
<body>
  <!-- Image to detect -->
  <img id="img" src="dog.jpg"/>
  <!-- Load your external JS file (AFTER tf and coco-ssd) -->
  <script src="P5_Starter_COCO_SSD.js"></script>
</body>
</html>
```

tfjs Pre-Trained models

- Object Detection (coco-ssd)

```
const img = document.getElementById('img');

// Wait until the image is fully loaded
img.onload = async () => {
  // Load the model
  const model = await cocoSsd.load();

  // Run detection
  const predictions = await model.detect(img);

  console.log('Predictions:', predictions);
};
```

tfjs Pre-Trained models

- Object Detection (coco-ssd)

```
const img = document.getElementById('img');
const canvas = document.getElementById('canvas');
const ctx = canvas.getContext('2d');

img.onload = async () => {
  // Match canvas size to image
  canvas.width = img.width;
  canvas.height = img.height;
```

tfjs Pre-Trained models

- Object Detection (coco-ssd)

```
// Draw each prediction
  predictions.forEach(prediction => {
    const [x, y, width, height] = prediction.bbox;
    // Draw rectangle
    ctx.strokeStyle = "red";
    ctx.lineWidth = 2;
    ctx.strokeRect(x, y, width, height);

    // Draw label background
    ctx.fillStyle = "red";
    ctx.fillRect(x, y - 20, width, 20);

    // Draw label text
    ctx.fillStyle = "white";
    ctx.font = "16px Arial";
    ctx.fillText(
      `${prediction.class} (${(prediction.score * 100).toFixed(1)}%)`,
      x + 4,
      y - 5
    );
  });
```

tfjs Pre-Trained models

- Common Objects in Context (COCO) dataset
- https://cocodataset.org/#home
- See paper on canvas that describes the dataset
  - COCO_dataset

Tensorflow.js App

- Can we train a model without programming?
  - Transfer Learning
    - Use relevant parts of a pre-trained ML model and apply it to a new but similar problem like classification
    - The pre-trained model will be the core information
    - To solve a new problem new data aspects will be added
  - https://teachablemachine.withgoogle.com/

Tensorflow.js App

- Teachable Machine
  - Transfer Learning Export
  - Teachable Machine Image Project
    - Teachable_Viewer1.0.html
    - Teachable_Viewer1.0.html
  - Teachable Machine Pose Project
    - Teachable_Preview_Pose.html
      - Needs some work
      - Possible extra credit

tfjs Pre-Trained models

- Pose Detection
- https://github.com/tensorflow/tfjs-models/blob/master/pose-detection/README.md

- PoseNet Demo (17 key points)
- https://storage.googleapis.com/tfjs-models/demos/pose-detection/index.html?model=posenet

- MoveNet Demo (17 key points)
- https://storage.googleapis.com/tfjs-models/demos/pose-detection/index.html?model=movenet

- BlazePose Demo (33 keypoints)
- https://storage.googleapis.com/tfjs-models/demos/pose-detection/index.html?model=blazepose

tfjs Pre-Trained models

- Hand Pose Detection
- https://github.com/tensorflow/tfjs-models/blob/master/hand-pose-detection/README.md

- MediaPipe Hands
- https://storage.googleapis.com/tfjs-models/demos/hand-pose-detection/index.html?model=mediapipe_hands

- Tracks 21 points per hand in 3D
- Detect Gestures
- Sign Language
- User interfaces similar to Mixed Reality Hololens2

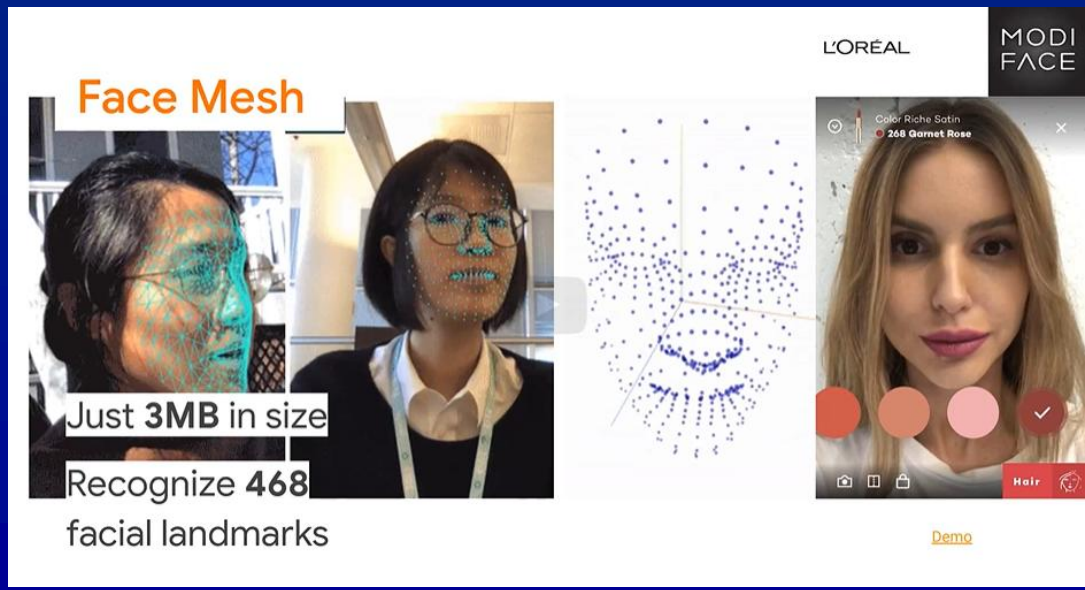
- MediaPipe Hands (Lite/Full)
- https://drive.google.com/file/d/1-rmIgTfuCbBPW_IFHkh3f0-U_lnGrWpg/view

tfjs Pre-Trained models

- Body Segmentation
- https://github.com/tensorflow/tfjs-models/blob/master/body-segmentation/README.md
- Distinguishes 24 body areas, across multiple bodies
- Body Pix Demo
- https://storage.googleapis.com/tfjs-models/demos/body-pix/index.html
- Body Pix Demo
- https://storage.googleapis.com/tfjs-models/demos/body-segmentation/index.html?model=selfie_segmentation

tfjs Pre-Trained models

- Face Mesh
- Face Landmarks Detection Models
- https://storage.googleapis.com/tfjs-models/demos/facemesh/index.html
- Integrates WebGL rendering
  - Irises are rendered in red
  - GLSL shaders!!



tfjs Pre-Trained models

- Modiface
- https://modiface.com/
- Leading provider of Augmented technology for the beauty industry
- Patent that includes:
  - Realtime 3D makeup simulation
  - Realtime anti-aging/skin-care simulation, and hair style and color simulation
  - Research from Stanford University and the University of Toronto

ModiFace

- Loreal
- https://www.loreal.com/en/articles/science-and-technology/l-oreal-modiface-brings-ai-powered-virtual-makeup-try-ons-to-amazon

CV App

- ETH white cane

https://www.greaterzuricharea.com/en/news/eth-white-cane-guides-using-camera-and-tactile-cues

CV App

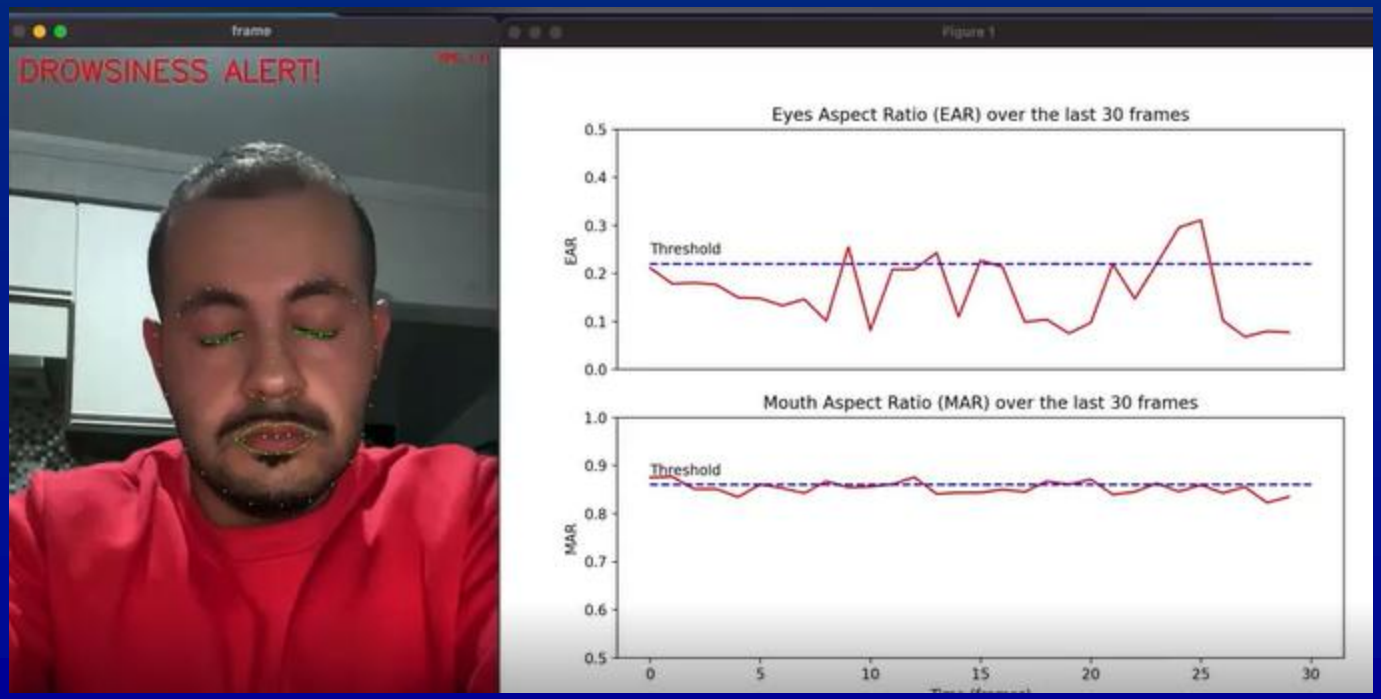
- Drowsiness Detection
- Thiago Silveira Machado
  - Drowsiness Detection is a technique that can be used in automobiles to reduce the number of traffic accidents caused by driver fatigue
  - In 2019, the National Highway Traffic Safety Administration (NHTSA) reported approximately 792 deaths due to drowsy driving.

CV App

- Drowsiness Detection
- Thiago Silveira Machado
  - According to F. Faraji et al. in "Drowsiness Detection Based On Driver Temporal Behavior Using a Newly Developed Dataset," this state of drowsiness can be observed through physiological measurements such as changes in brain waves and ECG, as well as vehicle-based measures, including:
    - - Lane deviation
      - Random steering movements
      - Changes in accelerator pedal pressure

CV App

- Drowsiness Detection (continued)
- Additionally, drowsiness can be detected through driver observation, including:
  - Increased blink frequency
  - Signs of drowsiness
  - Yawning
  - Head pitch changes
  - Lack of attention to the road

CV App

- Drowsiness Detection (continued)
  - The approach taken in this study aims to observe drowsiness and yawning states using computer vision techniques in Python, specifically utilizing dlib library for landmark detection
  - By detecting landmarks, we can determine the Eyes Aspect Ratio (EAR) to identify blink moments and evaluate driver's drowsiness, as well as the Mouth Aspect Ratio (MAR) to identify yawning moments.